10-K Similarity (Lazy Prices)

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April 6, 2026

1 Introduction

As one of the seven deadly sins slothfulness — or in other words, laziness — is one of the downfalls of humanity (according to Ponticus). Laziness is found in financial markets as well, as investors are unable or unwilling to incorporate information into financial markets; thus, creating market inefficiency.

This phenomenon is primary driver for the *Lazy Prices* trading strategy. Cohen, Malloy, and Nguyen (CMN) argue that because of the size of 10-K (and 10-Q) documents, investors are unable/unwilling to incorporate all of the relevant information into the stock; thus, creating a mispricing. CMN propose a strategy in their paper where one goes long in stocks that are non-changers and shorts changers in order to collect risk-neutral return.

They argue that firms that do not change their 10-K/10-Q filings as much will be comparatively less mispriced than non-changers (also the mispricing CMN find typically is a lack of incorporating bad information).

2 Data

2.1 CMN's Data

"We download all complete 10-K, 10-K405, 10-KSB and 10-Q filings from the SEC's Electronic Data Gathering, Analysis, and Retrieval (EDGAR) website from 1995 to 2014. All complete 10-K and 10-Q filings are in HTML text format and contain an aggregation of all information that are submitted with each firm's file, such as exhibits, graphics, XBRL files, PDF files, and Excel files. Similar to Loughran and McDonald (2011), we concentrate our analysis on the textual content of the document. We only extract the main 10-K and 10-Q texts in each document and remove all tables (if their numeric character content is greater than 15 XLS, and other binary files (CMN 2019)."

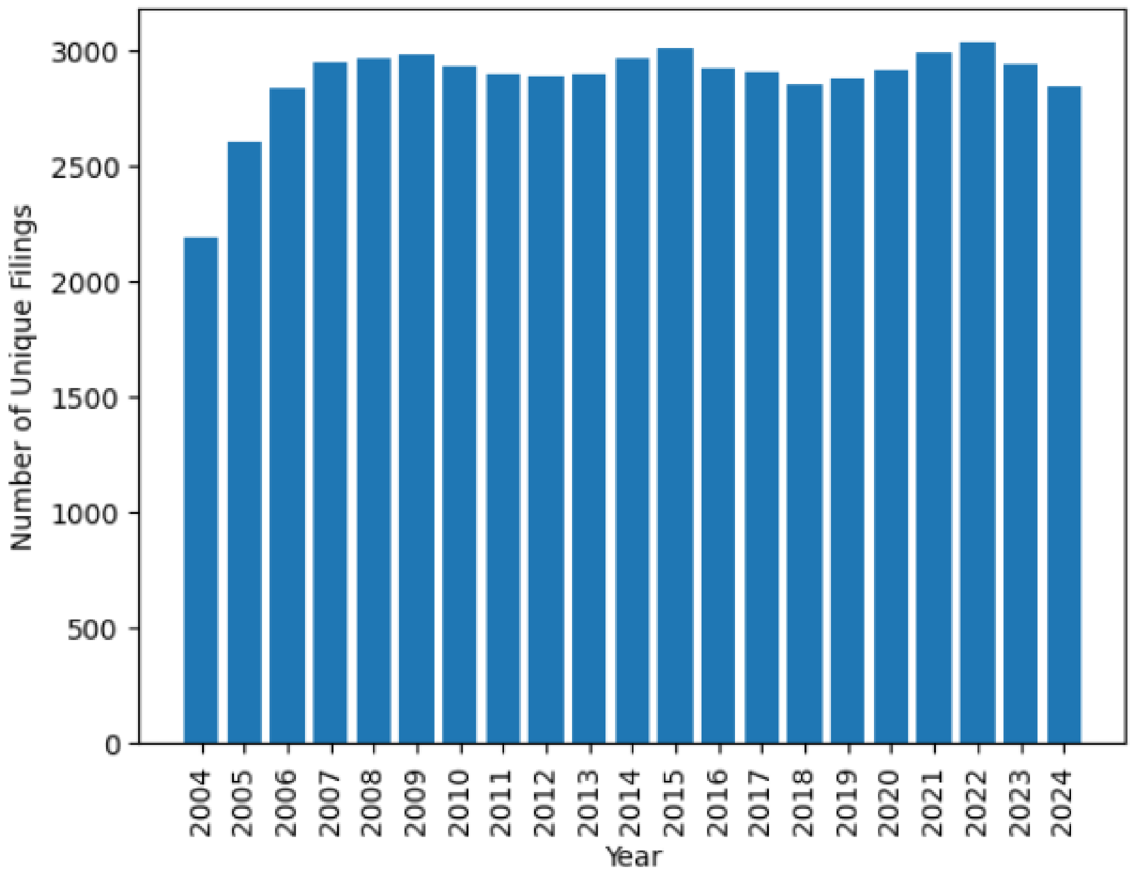


Figure 1: Russell-3000 10-K Coverage

2.2 Our Data

Similar to CMN's approach we use data from SEC EDGAR. To limit the amount of cleaning we had to do, we utilize the free API wrapper *edgartools* where all semi-recent ($year \geq 2004$) filings are cleaned in nice html/xrbl format.

Through this API, we use CIK [1] to query for each of the 10-K filings for stocks in our benchmark; doing so yields the coverage seen in Figure 1. In order to conduct our replication and extension we use the following data:

1. 10-K data (edgartools)

2. Pricing data (Barra/CRSP)

[1]CIK can be obtained through COMPUSTAT on WRDS; Compustat Capital IQ → Compustat North America → Fundamentals Annual

3 CMN's Results

CMN find that firms that make larger textual changes ("changers") significantly underperform firms that make few changes ("non-changers"), with a long–short portfolio earning roughly 34–58 basis points per month even after controlling for standard risk factors. Importantly, they find there is no essentially no announcement-day return; instead, returns accrue gradually over the following 6–12 months, consistent with investor inattention rather than risk compensation. Changes are especially predictive when concentrated in risk factors (Item 1A), litigation language (Item 3), and executive turnover references (Item 7), and they also forecast future negative earnings surprises, profitability declines, news events, and bankruptcies.[2]

4 Our Replication

Since, for our replication, we only had 10-K data (whereas CMN have 10-K and 10-Q data), we should expect our initial results to differ slightly (but perhaps remain directionally similar). Further, the data we were able to get from EDGAR was from 2004 to 2024. Thus, we have nearly 10 years in sample and 10 years out of sample — which may help us know if this signal/paper was simply data-mined.

4.1 Data Cleaning

While the data we got from *edgartools* was already quite clean, we did some initial cleaning[3] to make sure that we have the right year comparison for each CIK.

4.2 Similarity Calculation

CMN utilize four different similarity metrics to calculate similarity between 10-K filings and previous year - same quarter 10-Q filings. The four methods they use are: Cosine Similarity, Jaccard Similarity, MinEdit, Simple Similarity.[4]

$$CosineSim(A, B) = \frac{A \cdot B}{\|A\|\|B\|} \tag{1}$$

Cosine similarity is calculated by vectorizing the 10-K/10-Q text. This vectorization is done in such a way where each word is assigned a count value (how many times it appears

[2]This is my edited version of ChatGPT's summary; figured I would give it credit.
[3]Code for cleaning is available at the GitHub[3] link provided in the references.
[4]A much more detailed explanation of their similarity metrics is found in the data section of their paper.

in the text). After repeating this process for the both documents, the Cosine similarity of the two vectors is calculated as shown in (1).

In our replication, we only use Cosine similarity as it is inherently simple to understand and because in *Lazy Prices* it seems to be the best at picking up on differences in item text.

With that being said, in the future we plan to research other methods of calculating similarity — whether that be methods used in *Lazy Prices* or more novel methods (such as ML or DL approaches).

4.3 Portfolio Construction

In CMN's construction, they form quintile portfolios based on similarity — as such, we do the same in order to match their methodology. Furthermore, because CMN have 10-K and 10-Q data they define their holding period as 3 months — or in other words until their next data point. Since we do not have 10-Q data, we experiment with different holding periods but eventually decide on holding for 12 months[5] as it is most intuitive.

Additionally, CMN only reported five different items from 10-K/10-Q filings[6] While some analysis was done (on our part) to explore the similarity signal in other items present in 10-K filings, we spent very little time and energy researching this; thus, a logical extension for future research would be to explore other items more fully.

CMN (and us as well), find that `Item 1A - Risk Factors` is most predictive of future returns. Also it is important to note that we only did equal-weighting, whereas CMN did both equal and value weighting.[7]

Below we report our Fama-French 5-factor regression for the quintile sort of Item 1A; note that all other results for all items are reported in the appendix.

From the five factor regression below in Table 1., we see that the spread portfolio has a significant intercept of 65 bps (per month). Further, the spread portfolio does not load on any of the other risk factors, indicative of a truly risk neutral strategy.

While other items present in 10-K reports did not signal as well in our replication, Item 1A seems to be robust to lack of data (only having a yearly signal). One plausible explanation may be: firms often simply reuse the text of Item 1A in their 10-Q reports and then only annually update their risk factors in their corresponding 10-K.

With that being said, it seems that the alpha generated by most other 10-K/10-Q items

[5]CMN find some decay in the similarity signal over time. They estimate that this signal lasts approximately 6 months. Thus, one area of future research will be regarding the optimal holding period with 10-K data.

[6]The five items that CMN report are as follows: Item 1A - Risk Factors, Item 3 - Legal Proceedings, Item 7 - Management's Discussion, Item 7A - Market Risk Factors, Item 9B - Other Information.

[7]While CMN do both weighting schemes, it is important to note that most of the alpha reported was in the equal weight construction; perhaps indicative of the signal being more predictive in small-cap stocks.

are not robust to only having annual data. Thus, it may be useful to consider getting 10-Q data to fully take advantage of this signal in the future.

Table 1: Fama-French 5-Factor Regressions by Bin (Item 1A; 12-mo holding)

	Bin 0	Bin 1	Bin 2	Bin 3	Bin 4	Bin Spread
Intercept	-0.0026	0.0046	0.0066	0.0027	0.0039	0.0065
	(-1.12)	(1.97)	(2.69)	(1.44)	(2.31)	(2.37)
MKT	1.0654	1.0918	1.0838	1.1110	0.9985	-0.0670
	(19.43)	(19.87)	(18.75)	(25.33)	(24.94)	(-1.03)
SMB	0.6887	0.8226	0.6952	0.7678	0.8417	0.1530
	(7.06)	(8.42)	(6.77)	(9.84)	(11.83)	(1.33)
HML	0.1775	0.2252	0.2043	0.0701	-0.0282	-0.2057
	(1.83)	(2.32)	(2.00)	(0.91)	(-0.40)	(-1.80)
RMW	-0.0794	-0.0315	-0.2174	-0.1140	-0.2435	-0.1641
	(-0.65)	(-0.26)	(-1.68)	(-1.16)	(-2.71)	(-1.13)
CMA	0.0263	0.0500	-0.0667	0.0096	-0.0670	-0.0933
	(0.17)	(0.32)	(-0.40)	(0.08)	(-0.58)	(-0.50)

5 MVE Backtest

Motivated by the signifcant alpha in our five-factor regression for Item 1A, we attempt to do a MVE backtest in which we compute the Mean-Variance-Efficient portfolio for each trading day.

We use Grinold and Kahn's alpha formula (Equation 2). Where we ex-ante use an IC of 0.05.

$$alpha = IC * spec\_risk * z\_score \tag{2}$$

In order to translate our yearly signal into a 12-month holding, we forward fill our computed alpha 250 trading days (after lagging one day to avoid lookahead-bias). We fill the remaining null alpha values with zero.

After constructing the alphas as above, we run a MVE backtest using the Zero Beta and Zero Investment constraints. Using these constraints yields the active portfolio with with weights summing to zero. We also gamma-tune and use a gamma value of 4200 to get to $\approx 5\%$ active risk.

A plot of the MVE backtest along with its statistics are shown below; turnover, drawdown, leverage are shown in the appendix.

Figure 2: Item 1A MVE Backtest (12-mo holding)

	Count	Mean Return (%)	Volatility (%)	Total Return (%)	Sharpe
Portfolio	5285	4.79	5.13	165.64	0.93

Table 2: Item 1A MVE Backtest Summary Statistics

6 Tonal Analysis

[8] To establish a good baseline for tonal analysis and also generate labels for distillation, I used FinBERT-tone, a BERT-based model fine-tuned for financial sentiment analysis. Using this model, I obtained labels and sentiment scores (logits) for every item in every 10-K in 2014.

Distillation is a technique where a student model is trained to mimic the predictions of a teacher model. Utilizing FinBERT-tone as the teacher, I fine-tuned Llama-3.1-8B-Instruct using LoRA to classify sentiment of 10-K text. The final result was a trained Llama model that could mimic the final layer output (logits) of FinBERT (I refer to this as the soft label).

This model was trained as a sequence classification model using KL divergence to match the teacher's distribution. This setup trained LoRA adapters on the attention projection layers (q_proj, k_proj, v_proj, o_proj) with rank of 32, lora_alpha=32, and dropout. This

[8] This was my CS-474 project.

resulted in 27.3 million trainable parameters (0.36% of the base model). Training loss decreased from 14.92 at the first step to 3.42 at the final step.

This model was fine-tuned for 3 epochs with a learning rate of 1e-4 using the AdamW optimizer. The model was trained on 90% of the data and validated on the other 10%; fine-tuning was done on a NVIDIA A100 GPU hosted on Google Colab.

As one who is not the most adept with regards to finetuning, my goal was simply to test the model I developed for my CS-474 project and nothing more. In the case my model performed poorly, I was not willing to spend more time refining my model to attempt to get it to work.

Thus, to test my model I simply ran the deployed the model on the 10-K data available and then observed the results for items[9]: 1A, 7, and 7A. The results were subpar at best and thus I did not continue to research this avenue (results in Appendix).

7 Optimal Holding Period

To examine the optimal holding period, we use the same process as above to create spread portfolios; however, we do so for each $t \in \{1, 2, 3, ..., 12\}$ where t is the number of months we hold a stock in a given bin.

In Figure 3, we see that the 12-month holding spread (our original holding period), preforms quite well. However, it seems to be eclipsed by the 6,7,8, and 11 month holdings periods.

First, this may be indicative of the same pattern in the paper when they make mention of the alpha lasting approximately 6 months.

Next, the 11-month holding period spread consistently performs better than the 12-month holding spread. This is likely due to the 10-K being released during the 12th month and thus inherently changing the signal while we hold the stock until month-end. So by changing to 11-month holding, we can correct for this. This is showcased in Figure 26 and Table 18, as we utilize an MVE backtest with a 230 trading day holding period (approximately 11 months).

8 Price Filter Sensitivity

We run an MVE backtest filtering out stocks that have lagged price of less than 5 dollars and a non-null cosine similarity value. This effectively only removes observations where the

[9]Item 1A, 7, and 7A are the most discretionary; thus, a tonal analysis on these items is likely more viable than other items.

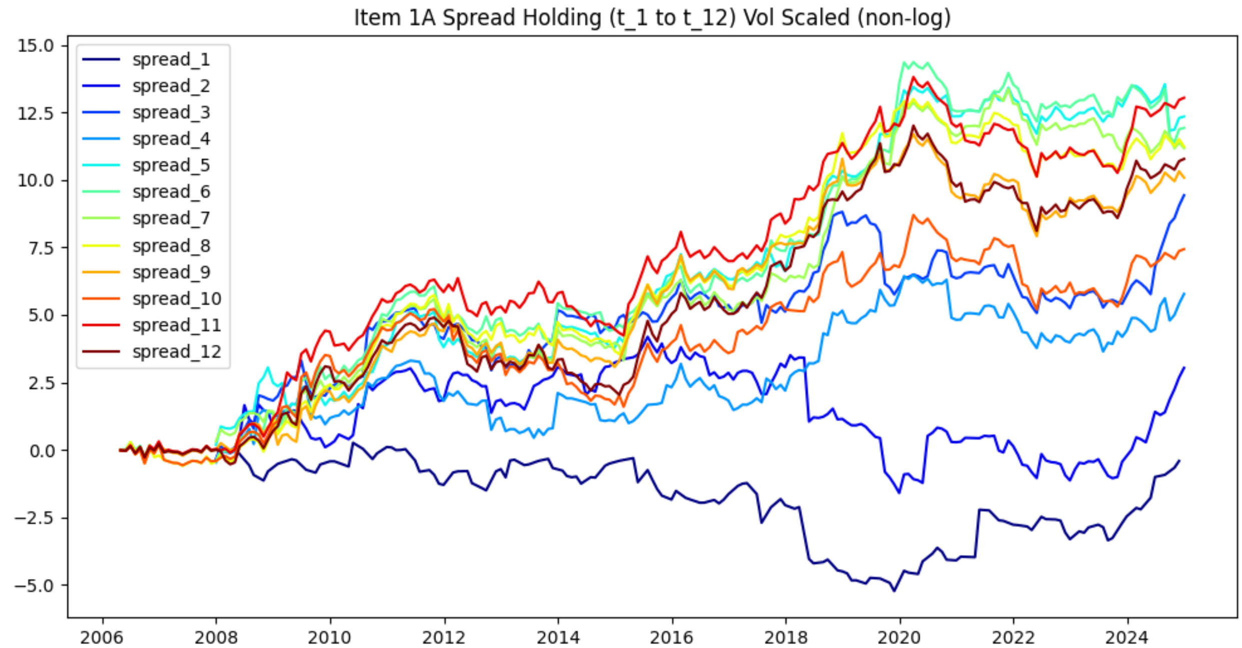


Figure 3: Spread Portfolios with Different Holdings

`prc_lag` is less than five the day of filing. Doing so allows us to hold stocks that dip below 5 dollars, but were not below five dollars at the time of purchase.

We find that a price filter (of this manner) has essentially no effect on the MVE backtest as we achieve the same results as in Figure 26 and Table 18.

9 Exploring Different Similarity Metrics

In CMN's original paper they employ 4 different similarity metrics (MinEdit, Simple, Jaccard, Cosine), finding cosine similarity to perform the best in their sample. However, as there is little to no economic intuition why cosine similarity should perform better than any other given similarity or distance metric, I hope to test a more complete set of distance and similarity metrics using many different methods.

I judge the similarity metrics based off their 11-mo holding spread portfolio Sharpe ratio (which may not be the best way to do it, but it's what I did). The Sharpe ratios for the different similarity metrics are reported in Table 3.

For vector-based similarity methods, I employ different vectorization methods as well which include: count, TF-IDF, and SBERT.[10] The count vectorizer[11] takes each word in

[10] I include vol-scaled plots of all methods tested over time in the Appendix.
[11] CMN only use count vectorizer when they do cosine similarity.

Item 1A and assigns each word a value based on the word count. TF-IDF is very similar; however, it down-weights common words and up-weights words that rarely occur between the documents. Lastly, SBERT is transformer-based method that takes the document and encodes it to a vector embedding using a neural network architecture.

Vectorizer / Method	**Metric**	**Sharpe**
Count Vectorizer	Cosine Similarity	0.485
Count Vectorizer	Pearson Correlation	0.475
Count Vectorizer	Euclidean Distance	-0.051
Count Vectorizer	Manhattan Distance	-0.223
Count Vectorizer	RBF Kernel	0.254
TF-IDF	Cosine Similarity	0.516
TF-IDF	Pearson Correlation	0.508
TF-IDF	Euclidean Distance	0.574
TF-IDF	Manhattan Distance	0.531
TF-IDF	RBF Kernel	0.545
SBERT	Cosine Similarity	0.383
SBERT	Pearson Correlation	0.381
SBERT	Euclidean Distance	0.430
SBERT	Manhattan Distance	0.472
SBERT	RBF Kernel	0.377
Distributional Distance (Count)	KL-Divergence	0.602
Distributional Distance (Count)	Jensen–Shannon divergence	0.530
Distributional Distance (Count)	Hellinger Distance	0.518
Jaccard	Jaccard	0.437
Jaccard	Weighted Jaccard	0.490

Table 3: Sharpe ratios for similarity and divergence metrics across vectorization methods

There are essentially three different methods employed in Table 3. First, vector-based methods; second, distribution based methods (where I use a count vectorizer to get a quasi probability of that word occuring). Lastly, I also use Jaccard-based methods which work more with sets rather than vectors or distributions.

In Table 3, there are a couple things worth noting. First, more complex methods such as SBERT perform poorly compared to other methods perhaps indicating the signal is much more the exclusion or inclusion of risk factors rather than context and meaning.

$$P_i = \frac{c_i + \epsilon}{\sum_j (c_j + \epsilon)}, \qquad Q_i = \frac{d_i + \epsilon}{\sum_j (d_j + \epsilon)} \tag{3}$$

$$D_{KL}(P \parallel Q) = \sum_i P_i \log\left(\frac{P_i}{Q_i}\right) \tag{4}$$

Further, it seems like methods that up-weight outliers seem to perform better (TF-IDF based methods and my implementation of KL Divergence); which also seems to be indicative of inclusion or exclusion of uncommon words being the driving force behind the signal.

To test/illustrate this further, I ran a MVE Backtest with KL-Divergence as defined in Equations 3 and 4, which achieved a Sharpe ratio of 1.08!

	Count	Mean Return (%)	Volatility (%)	Total Return (%)	Sharpe
Portfolio	5285	5.35	4.94	199.39	1.08

Table 4: 11-month KL Divergence MVE Backtest Summary Statistics

Term	Return
Intercept	0.0002 (3.95)
MKT	-0.0159 (-4.04)
SMB	0.0052 (0.72)
HML	-0.0058 (-0.91)
RMW	-0.0686 (-6.73)
CMA	-0.0300 (-2.33)

Table 5: KL-Divergence MVE Backtest FF5 Regression

10 Alpha Weighting

Because of the varying signal strength by relative day after the 10-k filing is made public. We attempt to "re-weight" our alphas depending on the day relative to filing for each individual security.

To go about creating a "weighting" function, we take our KL-Divergence backtest and fit a Fama-French 5 factor regression on our return time series (in-sample only). Afterwards, motivated by an extremely small $\mathbf{R^2}$, we take only the alpha and the average residual for each day relative to 10-K filing. We then fit a cubic spline to the signal in our predetermined in-sample period (2004-2020).

We use the fitted spline in the out of sample period (2020-2024) to test the validity of the doing the re-weighting scheme. We compare the weighted alpha MVE backtest to the baseline KL Divergence MVE backtest.

Table 6: Performance Comparison of Alpha Specifications

Metric	Weighted Alphas	Original Alphas
Sharpe Ratio	1.01	**1.34**

As we can see in Table 6, our weighting scheme seems to perform worse than simply forward filling the alphas. This may be indicative that the signal is dynamic over time or that we have little success using past data to predict signal strength over day.

Thus, it may be best to simply retain the forward filling function (while it is not the most intuitive, it performs better than any other specification). Note further graphs and tables are included in the Appendix.

11 Using Stop Words

Because a lot of words in 10-K filings contain little to no relevant information, I employ what is called a *stop words list*. This essentially is a list of words that— if found, will be removed from a set of text. Lists typically contain words such as he, she, it, they, the, and, etc.

While some words may seem more informative than others, this fact is completely context dependent. Some lists may improve performance and some may hinder it.[2] One may be able to significantly improve the strategy's performance by finding a suitable stop words list.

Because creating my own list would be arduous, I opted for the simplicity of using *sklearn's* own English stop words list. By using this list when we vectorize to create probability distributions for KL divergence, we are limiting to words that carry more information. Doing so yields nearly identical results but the Sharpe ratio increases to 1.09.

12 Momentum Analysis

The traditional intuition behind equity momentum is that is investors underact to new information and create the momentum effect. Since the momentum intuition is quite similar to the logic behind *Lazy Prices* (investors underreact and/or are lazy and slow at incorporating information from 10-K/10-Q into the market), I figured that I would check to see how adding momentum[12] as a factor in a factor regression changes our results.

As seen in Table 7, adding momentum (in the quintile sort), does not substantially affect the alpha that is observed there (this is consistent with CMN as they also found that the alpha was robust to momentum).

[12] Momentum data from Ken French's data library.

Table 7: Five Factor Regression with Momentum

Term	0	1	2	3	4	Spread
Intercept	-0.00 (-0.87)	0.01 (2.26)	0.01 (3.05)	0.00 (1.75)	0.00 (2.60)	0.01 (2.28)
MKT	1.01 (18.28)	1.04 (18.65)	1.02 (17.59)	1.07 (24.10)	0.96 (23.63)	-0.04 (-0.65)
SMB	0.66 (7.03)	0.80 (8.37)	0.67 (6.71)	0.75 (9.88)	0.82 (11.86)	0.16 (1.42)
HML	0.06 (0.56)	0.12 (1.22)	0.08 (0.77)	-0.02 (-0.28)	-0.10 (-1.41)	-0.16 (-1.30)
RMW	-0.09 (-0.76)	-0.04 (-0.33)	-0.23 (-1.82)	-0.12 (-1.28)	-0.25 (-2.86)	-0.16 (-1.10)
CMA	0.11 (0.73)	0.12 (0.78)	0.02 (0.12)	0.07 (0.60)	-0.01 (-0.13)	-0.13 (-0.67)
MOM	-0.20 (-3.77)	-0.17 (-3.14)	-0.20 (-3.61)	-0.15 (-3.53)	-0.12 (-3.11)	0.08 (1.19)

13 Next Steps

As alluded to earlier in the paper, there are many research avenues that could be worth exploring. Below, we list some areas that we are considering:

1. Look at different items.

2. Get 10Q data and reconstruct the paper more accurately.

3. Perhaps make a custom stop words list.

If I were to rank these by value, I would look into getting 10-Q data first and also exploring more of the items in 10-K/10-Q.

References

[1] Cohen, Lauren and Malloy, Christopher J. and Nguyen, Quoc, Lazy Prices (September 2018). NBER Working Paper No. w25084, Available at SSRN: https://ssrn.com/abstract=3254078

[2] Nothman, Joel and Qin, Hanmin and Yurchak, Roman, Stop Word Lists in Free Open-source Software Packages. In: Proceedings of the Workshop for NLP Open Source Software (NLP-OSS), Association for Computational Linguistics, Melbourne, Australia, July 2018, pp. 7–12. Available at: `https://aclanthology.org/W18-2502/`. doi: 10.18653/v1/W18-2502

[3] https://github.com/porterolson/lazy_prices

APPENDIX

Supplementary Quantile-based Tables/Charts

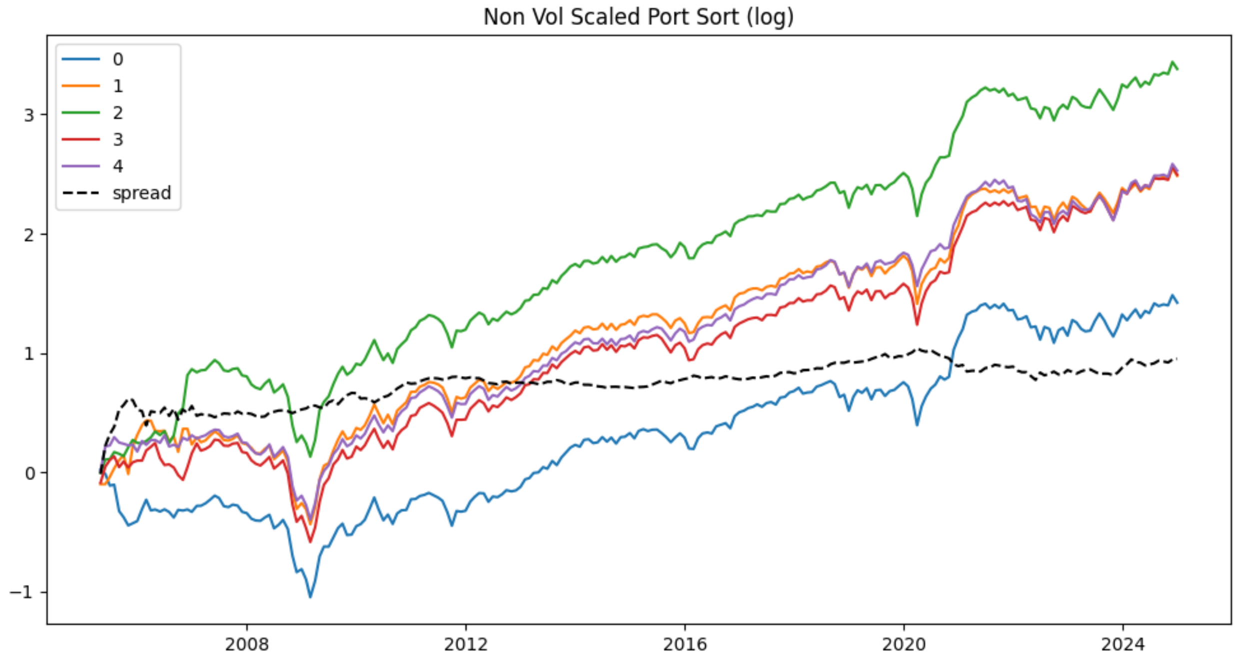


Figure 4: Raw Returns Item 1A Quintile Sort

Table 8: Fama-French 5-Factor Regressions by Bin (Item 1A; 12-mo holding)

	Bin 0	Bin 1	Bin 2	Bin 3	Bin 4	Bin Spread
Intercept	-0.0026	0.0046	0.0066	0.0027	0.0039	0.0065
	(-1.12)	(1.97)	(2.69)	(1.44)	(2.31)	(2.37)
MKT	1.0654	1.0918	1.0838	1.1110	0.9985	-0.0670
	(19.43)	(19.87)	(18.75)	(25.33)	(24.94)	(-1.03)
SMB	0.6887	0.8226	0.6952	0.7678	0.8417	0.1530
	(7.06)	(8.42)	(6.77)	(9.84)	(11.83)	(1.33)
HML	0.1775	0.2252	0.2043	0.0701	-0.0282	-0.2057
	(1.83)	(2.32)	(2.00)	(0.91)	(-0.40)	(-1.80)
RMW	-0.0794	-0.0315	-0.2174	-0.1140	-0.2435	-0.1641
	(-0.65)	(-0.26)	(-1.68)	(-1.16)	(-2.71)	(-1.13)
CMA	0.0263	0.0500	-0.0667	0.0096	-0.0670	-0.0933
	(0.17)	(0.32)	(-0.40)	(0.08)	(-0.58)	(-0.50)

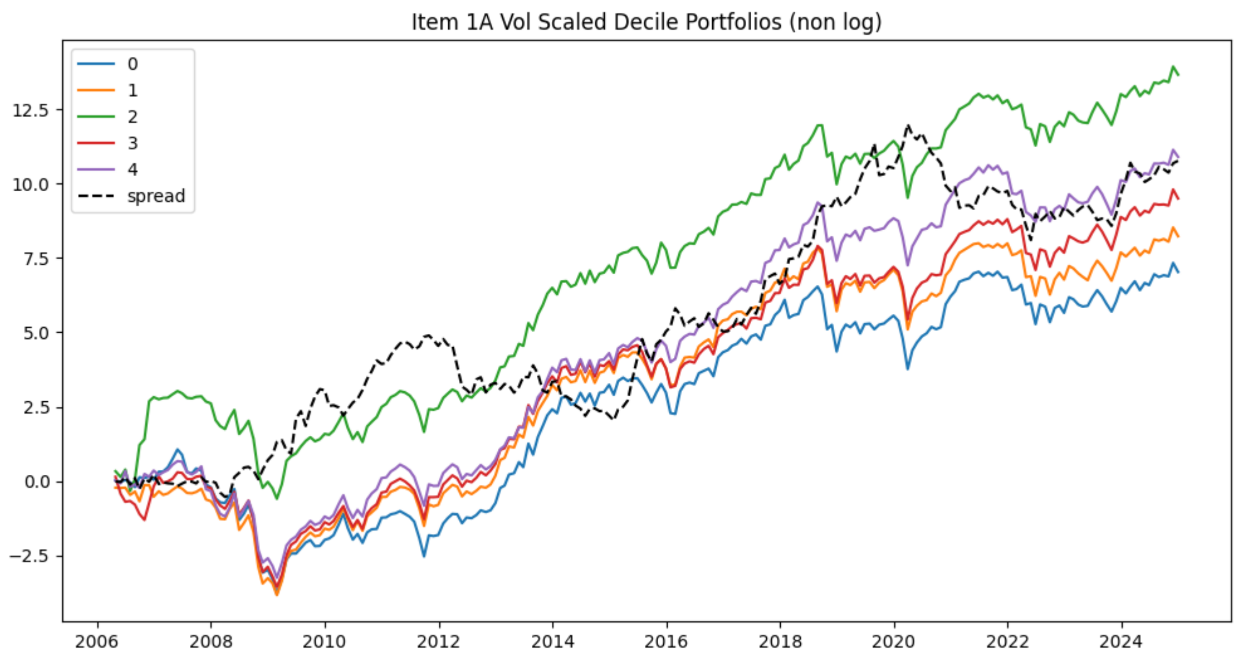


Figure 5: Volatility Scaled Returns: Item 1A

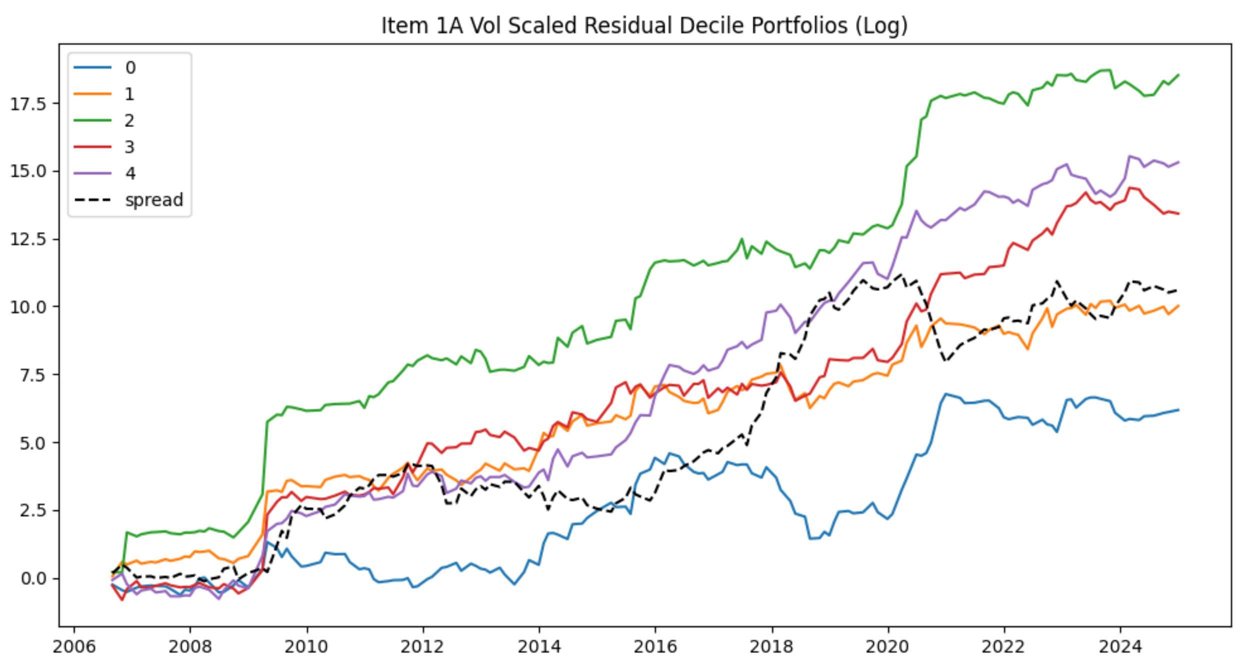


Figure 6: Residualized (to FF5) Returns for Item 1A (12-mo holding)

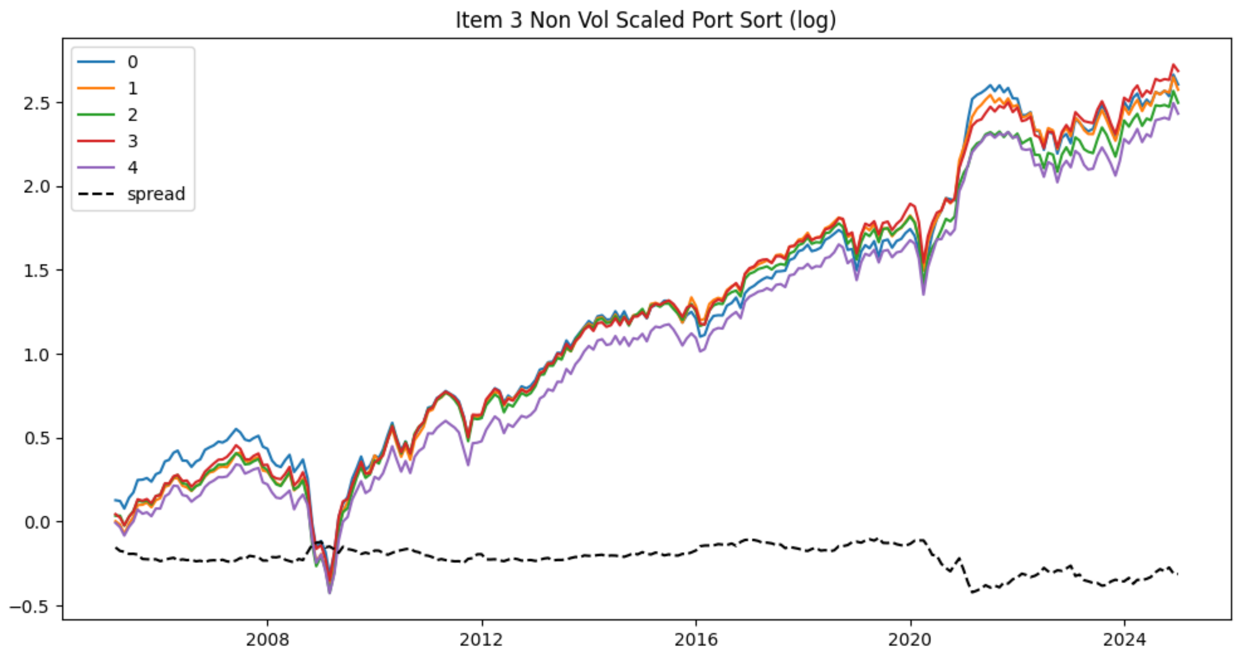


Figure 7: Raw Returns Item 3 Quintiles (12-mo holding)

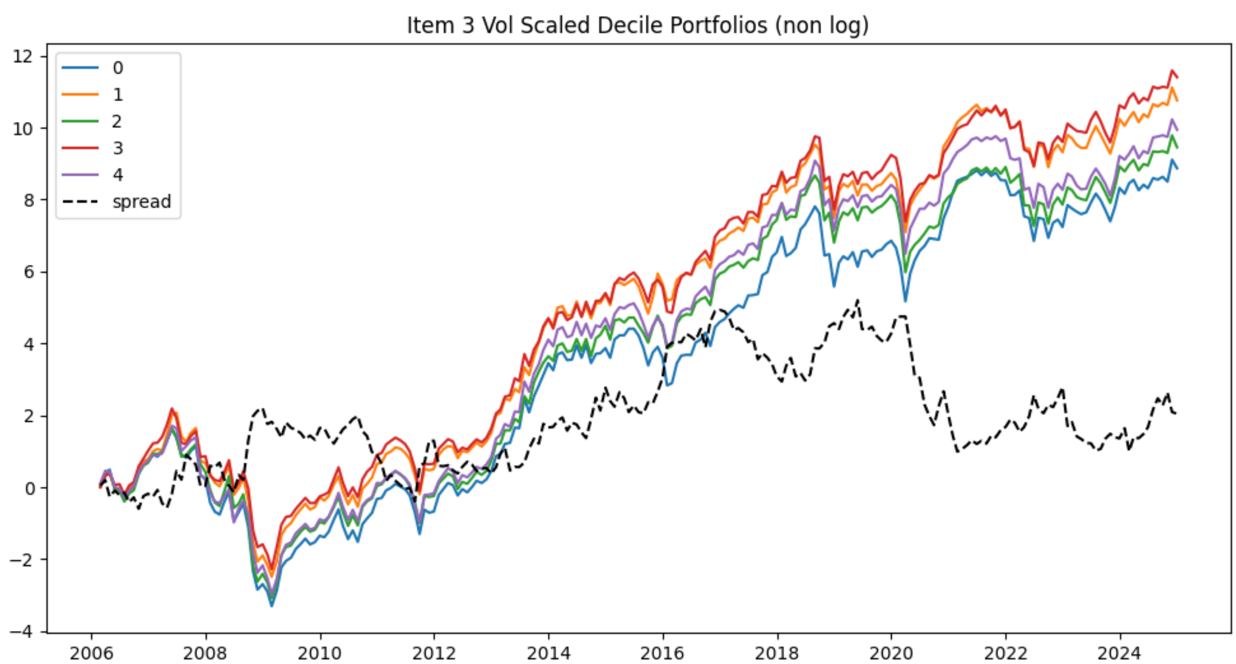


Figure 8: Vol Scaled Returns Item 3 Quintiles (12-mo holding)

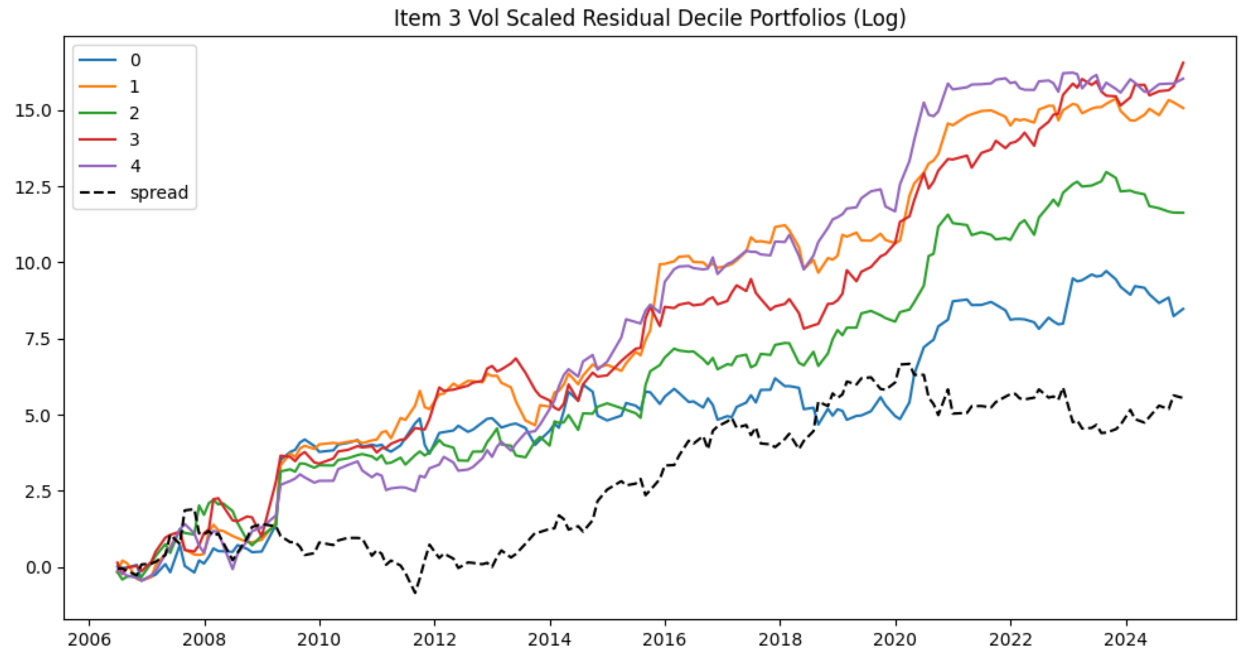


Figure 9: Residualized (to FF5) Returns for Item 3 (12-mo holding)

Table 9: Fama-French 5-Factor Regressions by Bin (Item 3)

	Bin 0	Bin 1	Bin 2	Bin 3	Bin 4	Bin Spread
Intercept	0.0031	0.0038	0.0029	0.0040	0.0031	0.0001
	(2.27)	(3.46)	(2.68)	(3.89)	(3.11)	(0.06)
MKT	1.1329	1.0815	1.0784	1.0311	0.9725	-0.1604
	(35.29)	(41.80)	(42.14)	(41.82)	(40.36)	(-4.63)
SMB	0.7511	0.8183	0.7567	0.7021	0.8336	0.0825
	(13.16)	(17.79)	(16.64)	(16.02)	(19.46)	(1.34)
HML	0.0742	0.0747	0.2584	0.1986	0.1776	0.1033
	(1.31)	(1.64)	(5.73)	(4.57)	(4.18)	(1.69)
RMW	-0.1932	-0.1349	-0.0713	-0.1792	-0.0958	0.0974
	(-2.68)	(-2.32)	(-1.24)	(-3.24)	(-1.77)	(1.25)
CMA	-0.0355	0.1005	-0.0673	-0.1338	-0.0299	0.0056
	(-0.38)	(1.35)	(-0.91)	(-1.88)	(-0.43)	(0.06)

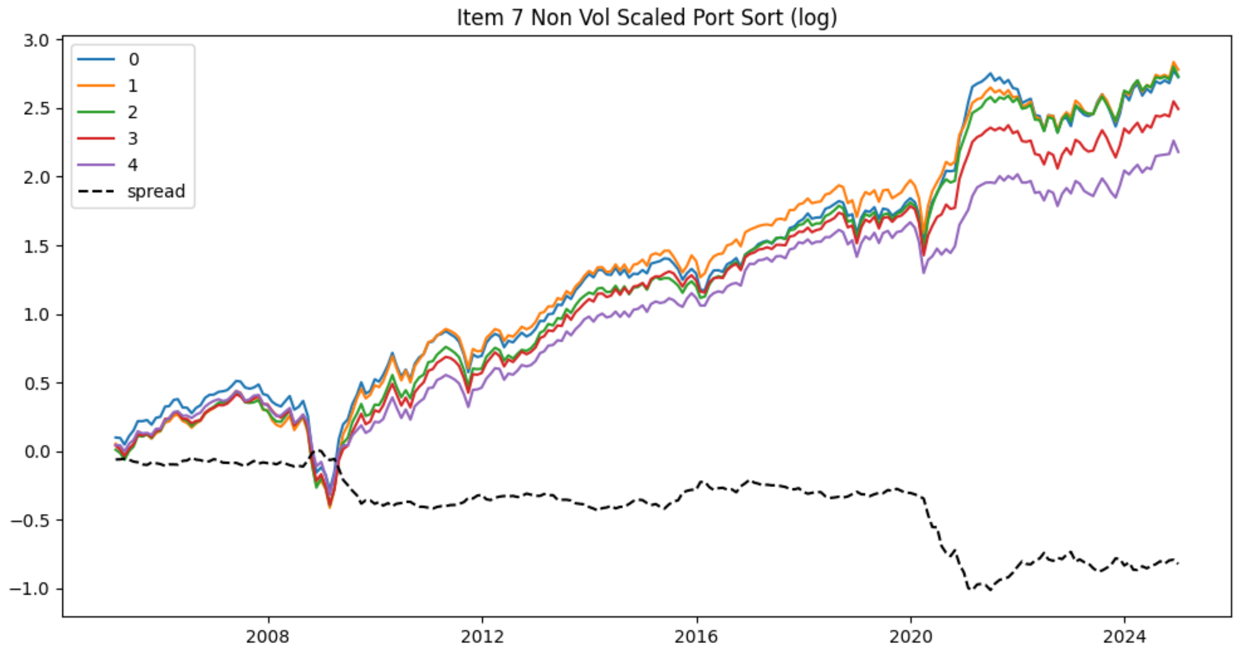


Figure 10: Raw Returns Item 7 Quintiles (12-mo holding)

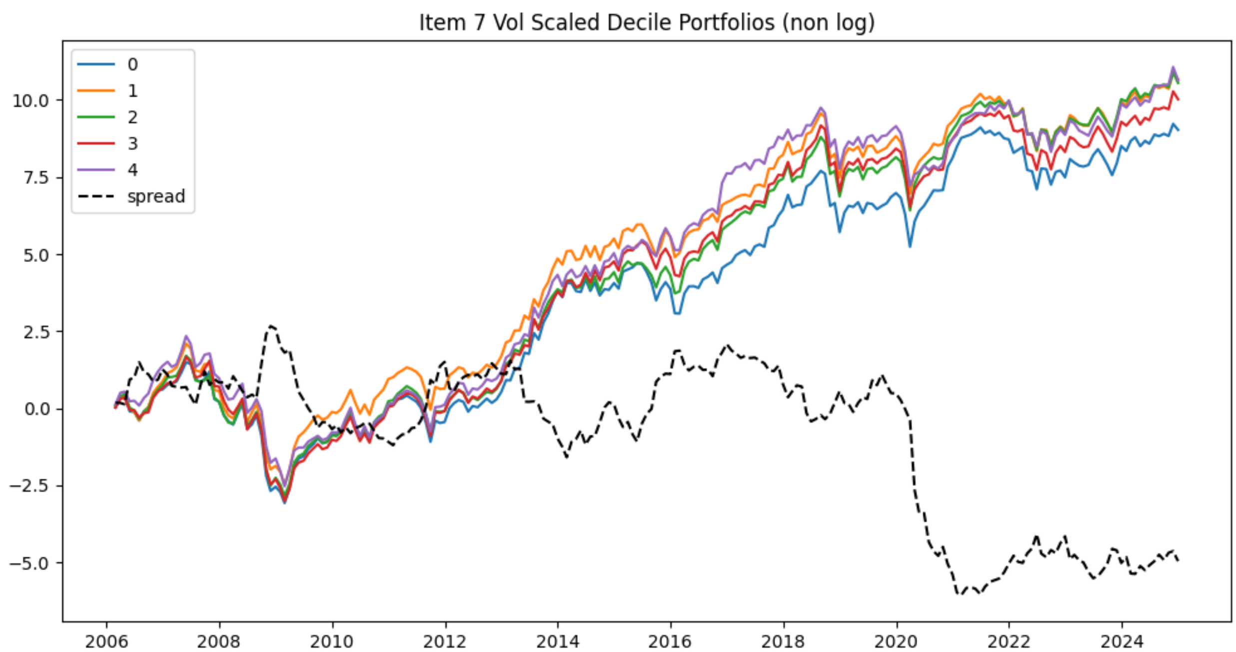


Figure 11: Vol-Scaled Returns Item 7 Quintiles (12-mo holding)

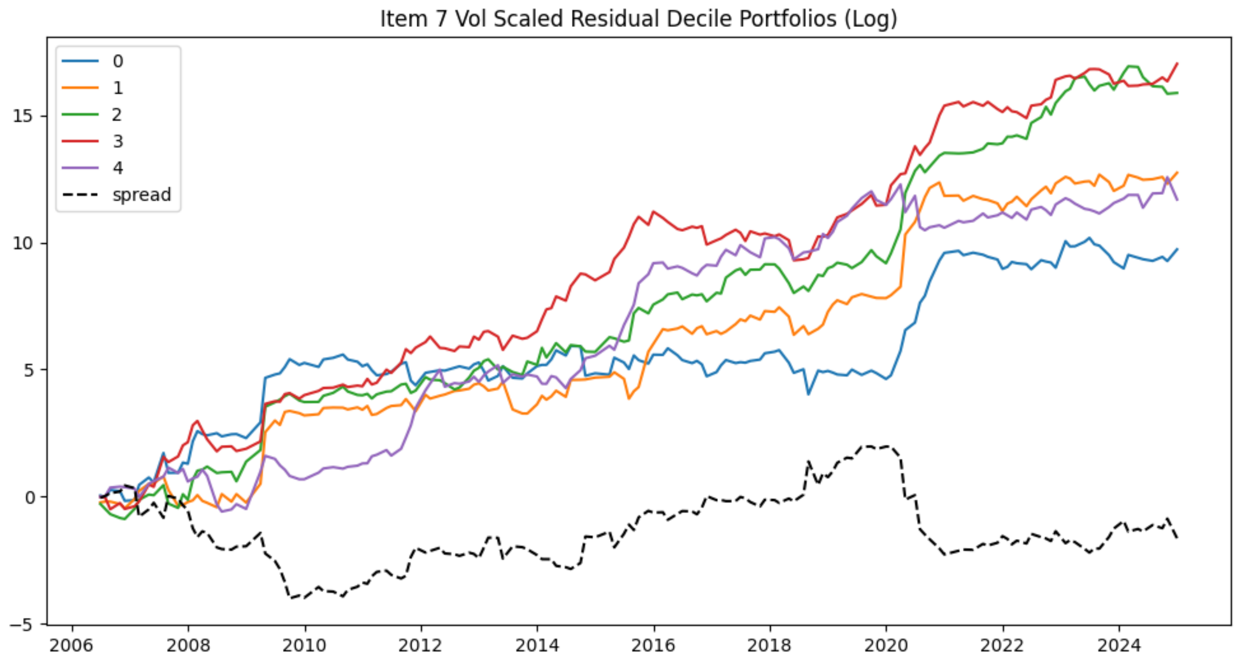


Figure 12: Residualized (to FF5) Returns for Item 7 (12-mo holding)

Table 10: Fama-French 5-Factor Regressions by Bin (Item 7)

	Bin 0	Bin 1	Bin 2	Bin 3	Bin 4	Bin Spread
Intercept	0.0037	0.0036	0.0043	0.0033	0.0026	-0.0011
	(2.65)	(2.90)	(3.84)	(3.87)	(2.50)	(-0.66)
MKT	1.1631	1.1173	1.0947	1.0418	0.8837	-0.2794
	(35.05)	(38.26)	(40.89)	(51.38)	(36.14)	(-6.92)
SMB	0.8166	0.8455	0.7965	0.7689	0.6406	-0.1760
	(13.84)	(16.29)	(16.74)	(21.33)	(14.74)	(-2.45)
HML	0.0457	0.1117	0.0794	0.1799	0.3570	0.3113
	(0.78)	(2.17)	(1.68)	(5.03)	(8.29)	(4.38)
RMW	-0.2217	-0.1460	-0.1797	-0.1219	-0.0193	0.2025
	(-2.98)	(-2.23)	(-2.99)	(-2.68)	(-0.35)	(2.24)
CMA	-0.0285	0.0210	0.0805	-0.0874	-0.1689	-0.1403
	(-0.30)	(0.25)	(1.04)	(-1.49)	(-2.39)	(-1.20)

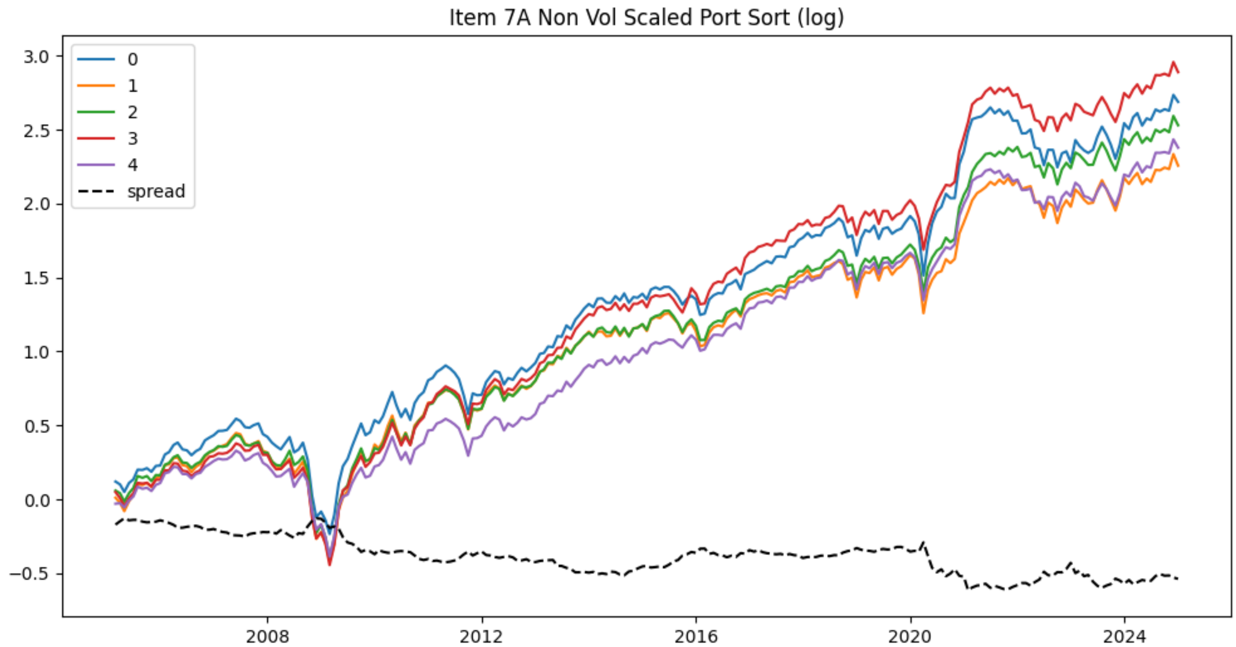


Figure 13: Raw Returns Item 7A Quintiles (12-mo holding)

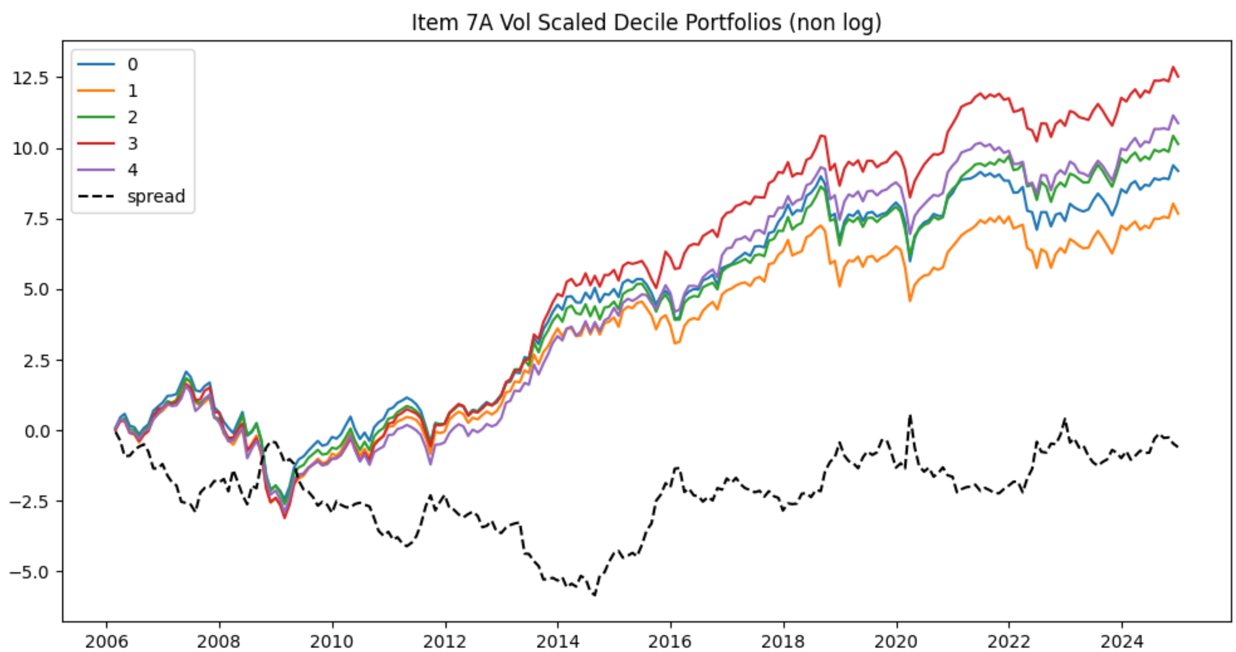


Figure 14: Vol-Scaled Returns Item 7A Quintiles (12-mo holding)

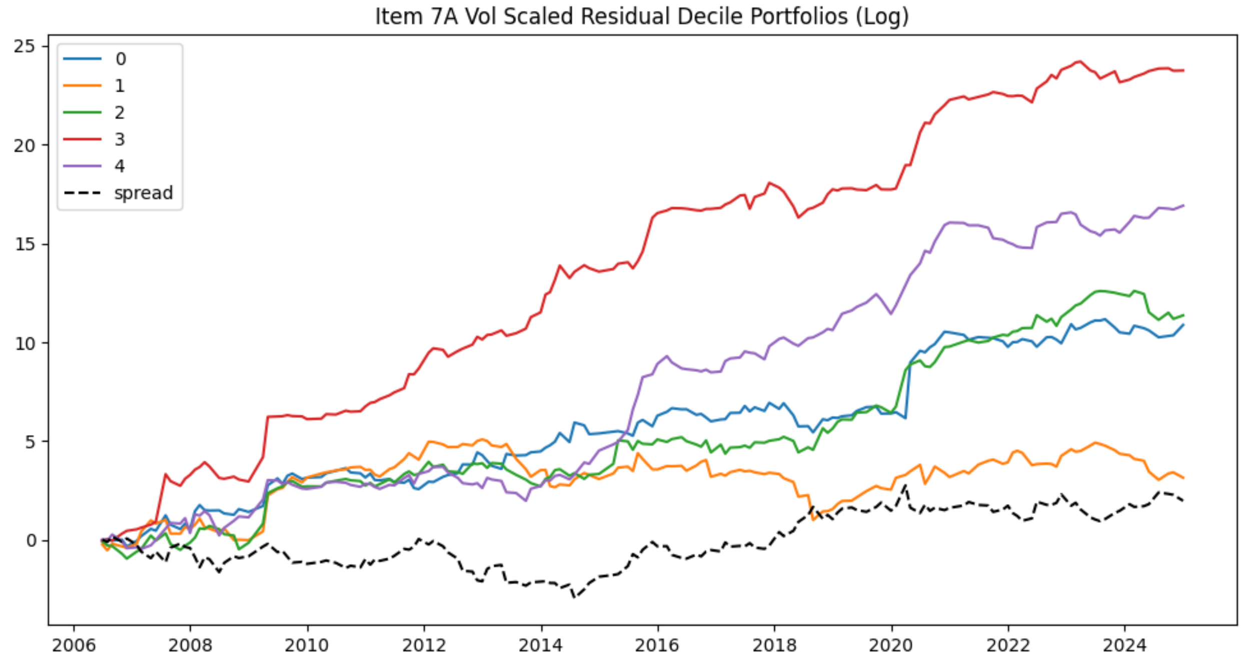


Figure 15: Residualized (to FF5) Returns for Item 7A (12-mo holding)

Table 11: Fama-French 5-Factor Regressions by Bin (Item 7A)

	Bin 0	Bin 1	Bin 2	Bin 3	Bin 4	Bin Spread
Intercept	0.0035	0.0018	0.0031	0.0049	0.0031	-0.0004
	(2.53)	(1.79)	(3.10)	(4.38)	(2.90)	(-0.26)
MKT	1.1645	1.0965	1.0861	1.0372	0.9206	-0.2439
	(35.37)	(46.33)	(45.98)	(38.83)	(36.22)	(-6.58)
SMB	0.7872	0.8206	0.7542	0.7788	0.7010	-0.0863
	(13.45)	(19.51)	(17.96)	(16.40)	(15.52)	(-1.31)
HML	0.1153	0.1618	0.1476	0.1521	0.2084	0.0931
	(1.99)	(3.88)	(3.55)	(3.23)	(4.65)	(1.43)
RMW	-0.1832	-0.0636	-0.0227	-0.2094	-0.1786	0.0046
	(-2.48)	(-1.20)	(-0.43)	(-3.50)	(-3.13)	(0.05)
CMA	-0.0981	0.0061	0.0506	-0.0916	-0.0230	0.0751
	(-1.03)	(0.09)	(0.74)	(-1.19)	(-0.31)	(0.70)

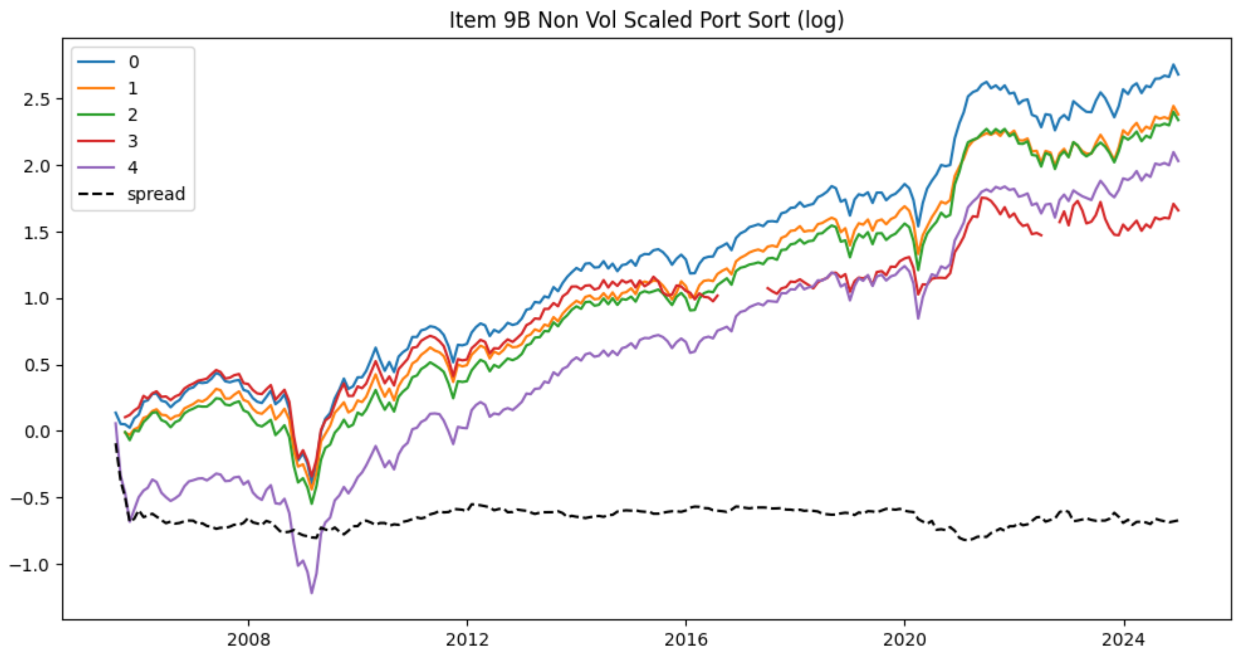


Figure 16: Raw Returns Item 9B Quintiles (12-mo holding)

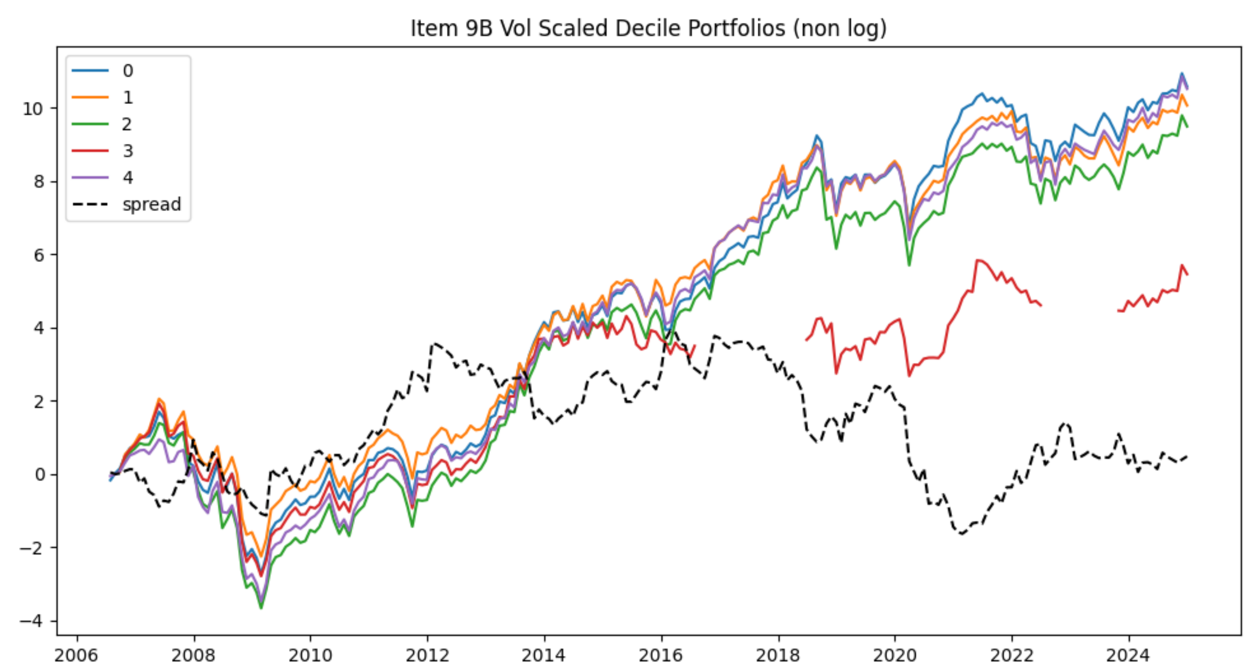


Figure 17: Vol-Scaled Returns Item 9B Quintiles (12-mo holding)

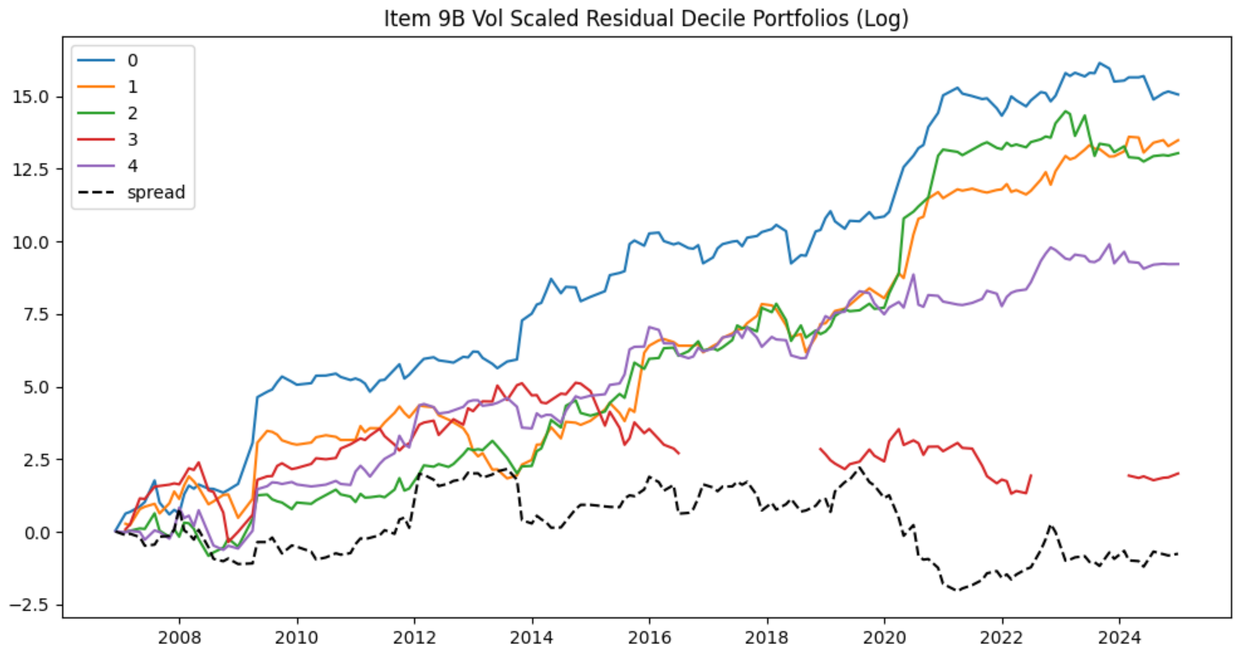


Figure 18: Residualized (to FF5) Returns for Item 9B (12-mo holding)

Table 12: Fama-French 5-Factor Regressions by Bin (Item 9B)

	Bin 0	Bin 1	Bin 2	Bin 3	Bin 4	Bin Spread
Intercept	0.0038	0.0029	0.0024	0.0024	-0.0002	-0.0039
	(3.06)	(2.68)	(2.23)	(1.01)	(-0.06)	(-1.57)
MKT	1.0974	1.0322	1.0461	0.8566	1.1263	0.0290
	(37.69)	(40.39)	(41.30)	(15.34)	(16.07)	(0.49)
SMB	0.8286	0.7689	0.8138	0.7673	0.8409	0.0123
	(15.94)	(16.84)	(17.98)	(7.67)	(6.72)	(0.12)
HML	0.0582	0.1627	0.0738	0.3496	0.1323	0.0740
	(1.13)	(3.59)	(1.64)	(3.53)	(1.07)	(0.71)
RMW	-0.1623	-0.0907	-0.1341	-0.1982	0.2147	0.3770
	(-2.49)	(-1.57)	(-2.35)	(-1.56)	(1.37)	(2.84)
CMA	-0.0114	0.0027	0.0294	-0.1601	0.0290	0.0404
	(-0.14)	(0.04)	(0.40)	(-1.00)	(0.14)	(0.24)

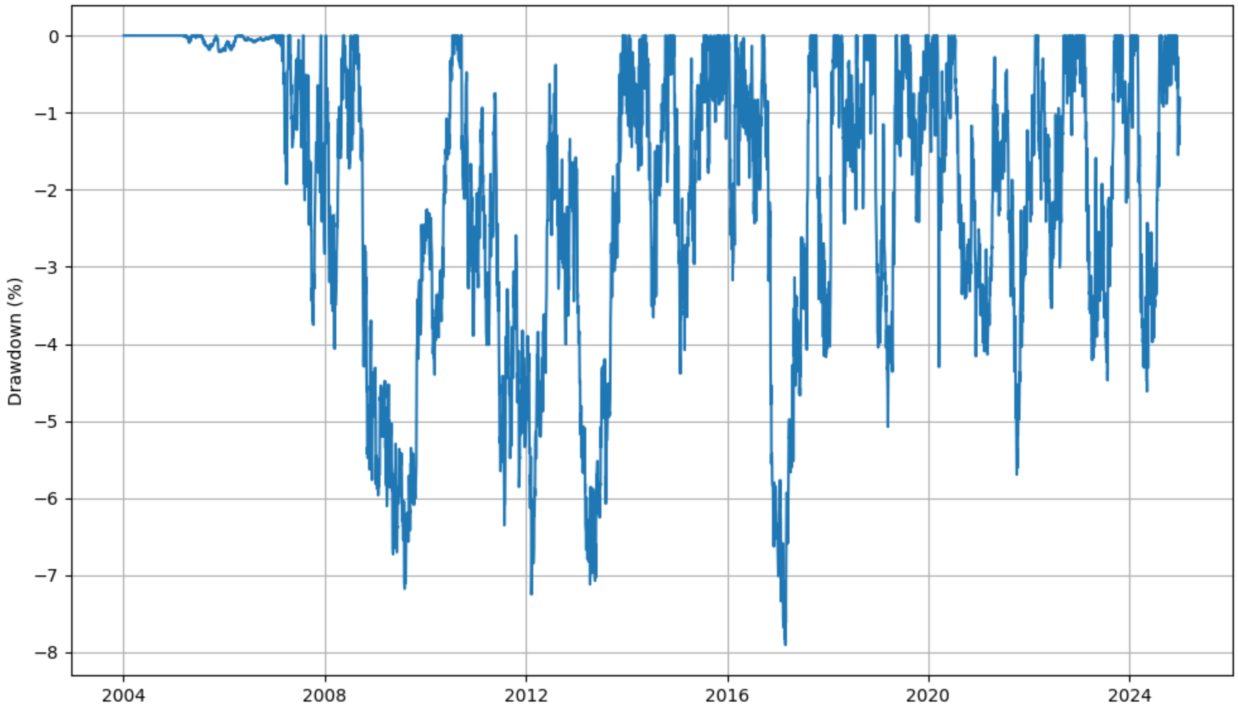
Supplementary MVE-Based Tables/Charts

Figure 19: Item 1A MVE Backtest Leverage (12-mo holding)

	Count	Mean Leverage	Min Leverage	Max Leverage	Std. Leverage
Portfolio	5285	4.83	0.00	7.67	2.38

Table 13: Leverage Summary Statistics

Figure 20: Item 1A MVE Backtest Drawdown (12-mo holding)



	Count	Mean (%)	Max (%)	Current (%)	Longest (days)
Porfolio	5285	-1.96	-7.90	-0.81	809

Table 14: Drawdown Statistics

Figure 21: Item 1A MVE Backtest Turnover (12-mo holding)

	Mean Turnover	Min Turnover	Max Turnover
Portfolio	0.1411	0.000	0.2502

Table 15: Turnover Statistics

Table 16: MVE Active Portfolio Fama-French 5 Factor Regression

	Active Portfolio
Intercept	0.0001
	(3.13)
MKT	-0.0146
	(-3.71)
SMB	0.0148
	(2.03)
HML	0.0033
	(0.51)
RMW	-0.0270
	(-2.66)
CMA	-0.0295
	(-2.31)

Figure 22: Tonal Quintile Sort (Vol-Scaled; 12-Month Holding)

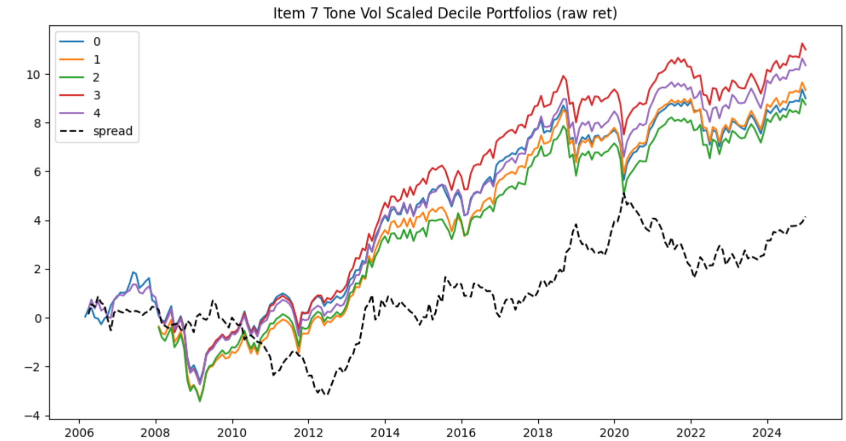


Figure 23: Item 1A Vol Scaled Tonal Quintile

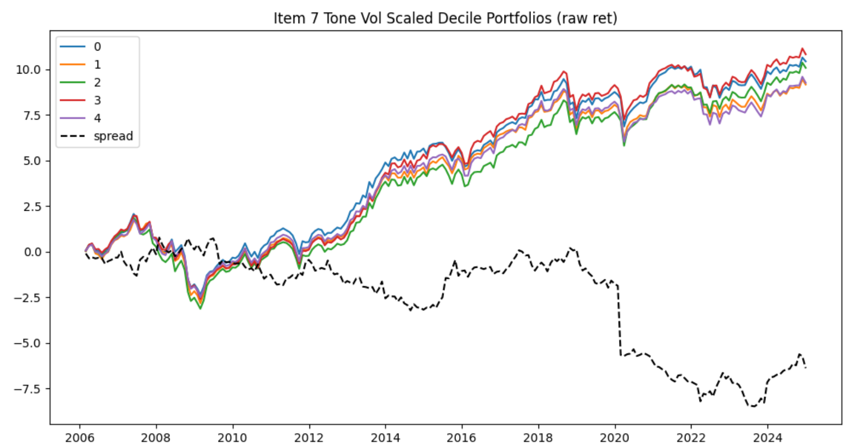


Figure 24: Item 7 Vol Scaled Tonal Quintile

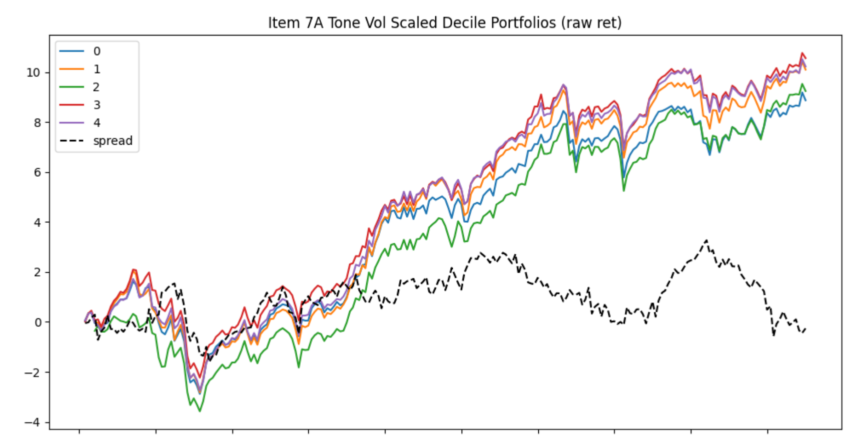


Figure 25: Item 7A Vol Scaled Tonal Quintile

Table 17: Fama-French 5-Factor Regressions by Bin

Item 1A Tonal Signal

	Bin 0	Bin 1	Bin 2	Bin 3	Bin 4	Spread
Intercept	0.0027	0.0031	0.0037	0.0040	0.0042	0.0015
	(2.89)	(3.46)	(2.92)	(3.94)	(2.84)	(1.05)
MKT	1.0864	1.0329	1.0933	1.0065	1.0180	-0.0683
	(48.80)	(51.13)	(38.12)	(43.45)	(29.10)	(-2.05)
SMB	0.7587	0.7742	0.7797	0.8047	0.7596	0.0008
	(19.17)	(20.92)	(14.84)	(18.97)	(12.22)	(0.01)
HML	0.2624	0.1676	0.2235	0.1795	0.1807	-0.0817
	(6.69)	(4.59)	(4.31)	(4.28)	(2.93)	(-1.39)
RMW	-0.0133	-0.1019	-0.0769	-0.1720	-0.3057	-0.2924
	(-0.27)	(-2.23)	(-1.19)	(-3.29)	(-3.90)	(-3.91)
CMA	-0.0743	-0.0322	-0.1132	-0.1261	-0.1036	-0.0292
	(-1.16)	(-0.54)	(-1.33)	(-1.84)	(-1.03)	(-0.30)

Item 7 Tonal Signal

	Bin 0	Bin 1	Bin 2	Bin 3	Bin 4	Spread
Intercept	0.0037	0.0033	0.0031	0.0032	0.0027	-0.0010
	(4.21)	(3.38)	(2.79)	(3.34)	(2.81)	(-1.17)
MKT	1.0926	1.0377	1.0765	1.0053	1.0316	-0.0610
	(52.06)	(44.08)	(40.28)	(44.10)	(44.90)	(-2.99)
SMB	0.7308	0.8033	0.8142	0.7268	0.7517	0.0208
	(19.59)	(19.20)	(17.14)	(17.94)	(18.41)	(0.57)
HML	0.2148	0.2012	0.1806	0.2004	0.2266	0.0118
	(5.81)	(4.85)	(3.84)	(4.99)	(5.60)	(0.33)
RMW	-0.1466	-0.1022	-0.1036	-0.1899	-0.0970	0.0496
	(-3.12)	(-1.94)	(-1.73)	(-3.72)	(-1.88)	(1.08)
CMA	-0.0655	-0.1152	-0.0556	-0.1126	-0.0846	-0.0191
	(-1.08)	(-1.70)	(-0.72)	(-1.71)	(-1.28)	(-0.32)

Item 7A Tonal Signal

	Bin 0	Bin 1	Bin 2	Bin 3	Bin 4	Spread
Intercept	0.0032	0.0034	0.0035	0.0029	0.0028	-0.0004
	(3.44)	(3.50)	(2.64)	(3.52)	(2.79)	(-0.41)
MKT	1.0156	0.9963	1.0487	1.0732	1.1204	0.1048
	(46.50)	(42.88)	(33.33)	(53.82)	(47.13)	(4.77)
SMB	0.7955	0.7785	0.7902	0.7355	0.7774	-0.0181
	(20.49)	(18.85)	(14.13)	(20.75)	(18.40)	(-0.46)
HML	0.1681	0.2573	0.2468	0.1747	0.1600	-0.0081
	(4.37)	(6.29)	(4.44)	(4.97)	(3.82)	(-0.21)
RMW	-0.1686	-0.1550	-0.1560	-0.0698	-0.0482	0.1205
	(-3.44)	(-2.98)	(-2.21)	(-1.56)	(-0.90)	(2.45)
CMA	-0.1090	-0.1020	-0.2346	-0.0426	0.0197	0.1287
	(-1.73)	(-1.52)	(-2.58)	(-0.74)	(0.29)	(2.03)

Figure 26: 11-month holding MVE Backtest

	Count	Mean Return (%)	Volatility (%)	Total Return (%)	Sharpe
Portfolio	5285	4.64	4.91	158.0	0.95

Table 18: 11-month MVE Backtest Summary Statistics

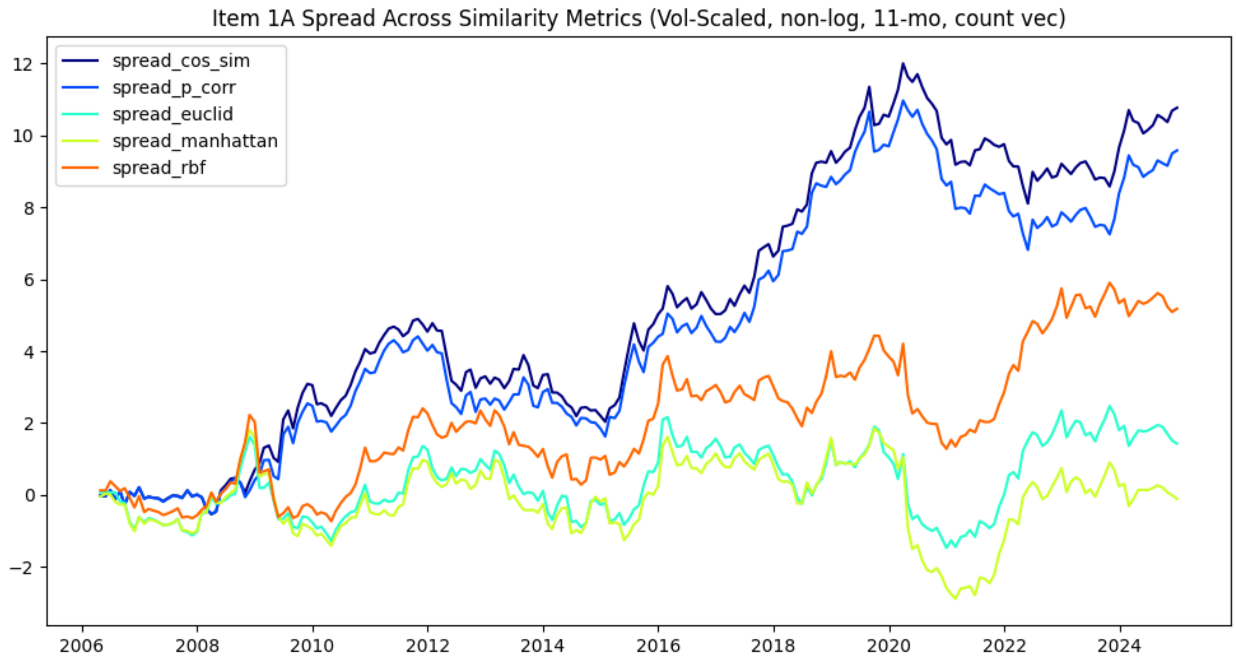


Figure 27: Count Vectorizer Based Similarity Metrics

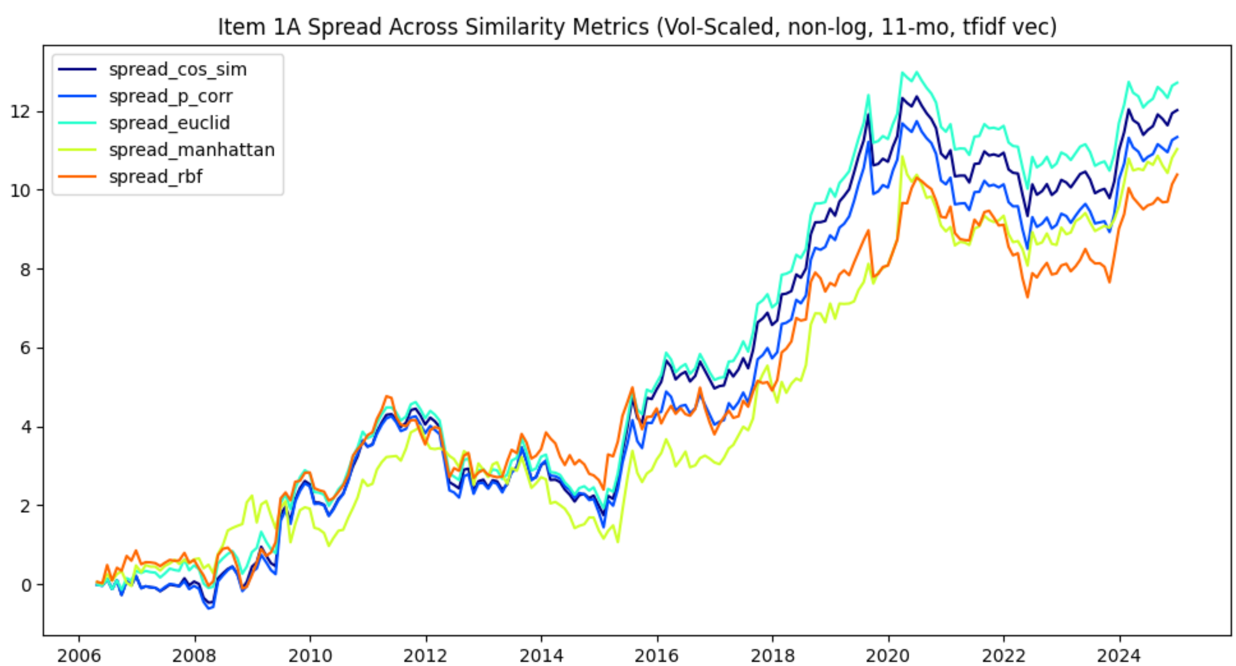


Figure 28: TF-IDF Vectorizer Based Similarity Metrics

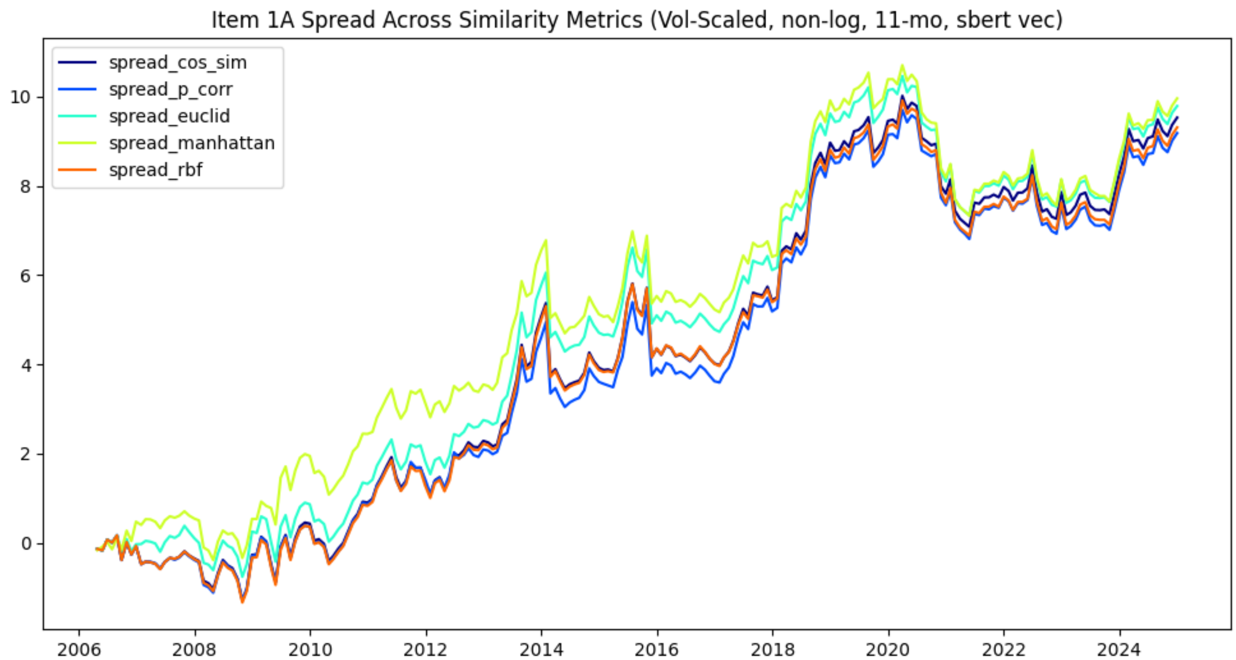


Figure 29: SBERT Vectorizer Based Similarity Metrics

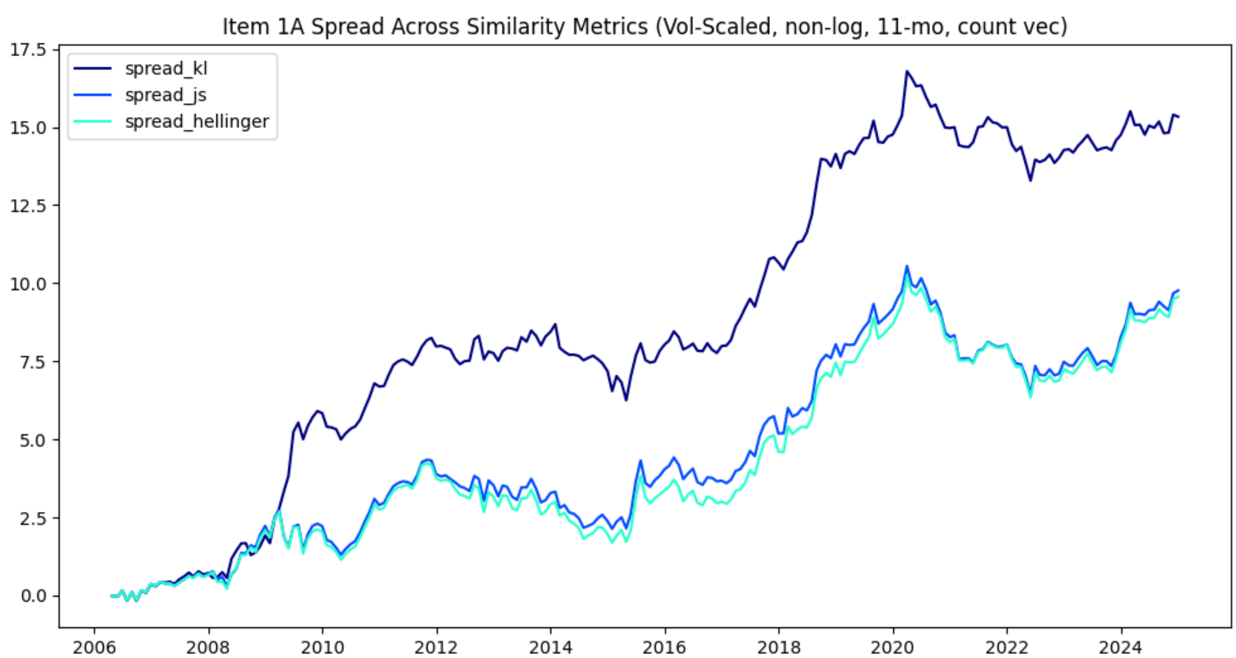


Figure 30: Distributional Based Similarity Metrics

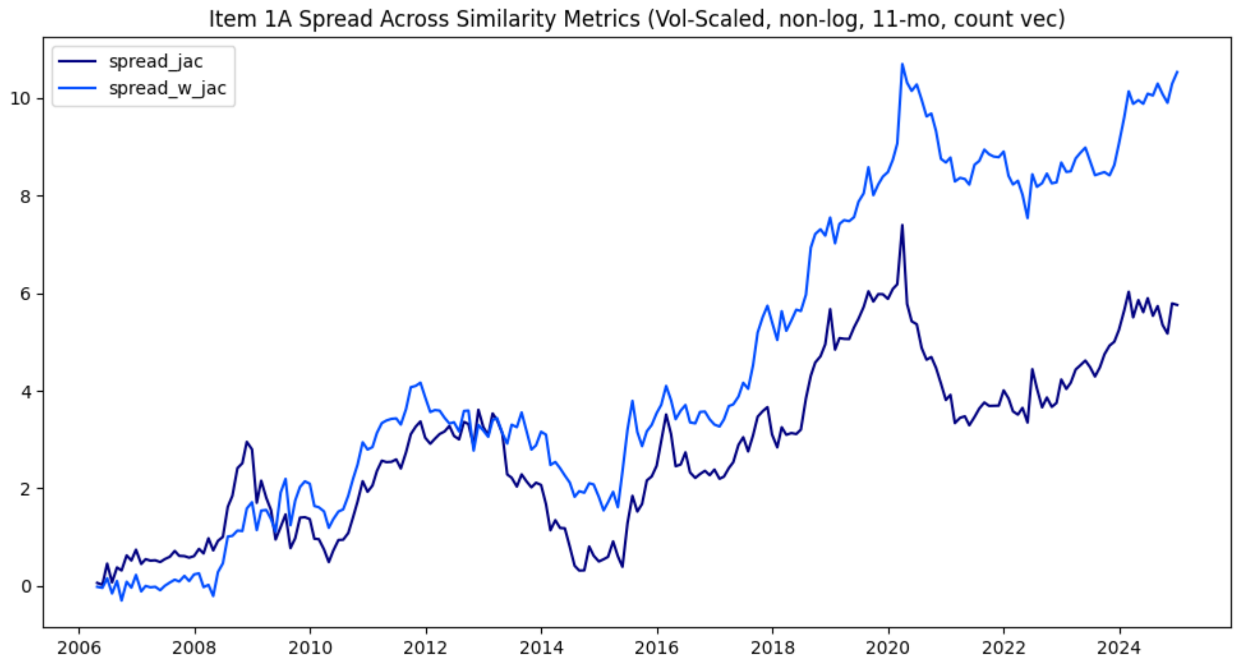


Figure 31: Jaccard & Weighted Jaccard (Using Count Vectorizer for weights)

Figure 32: KL-Divergence MVE Backtest (11-mo)

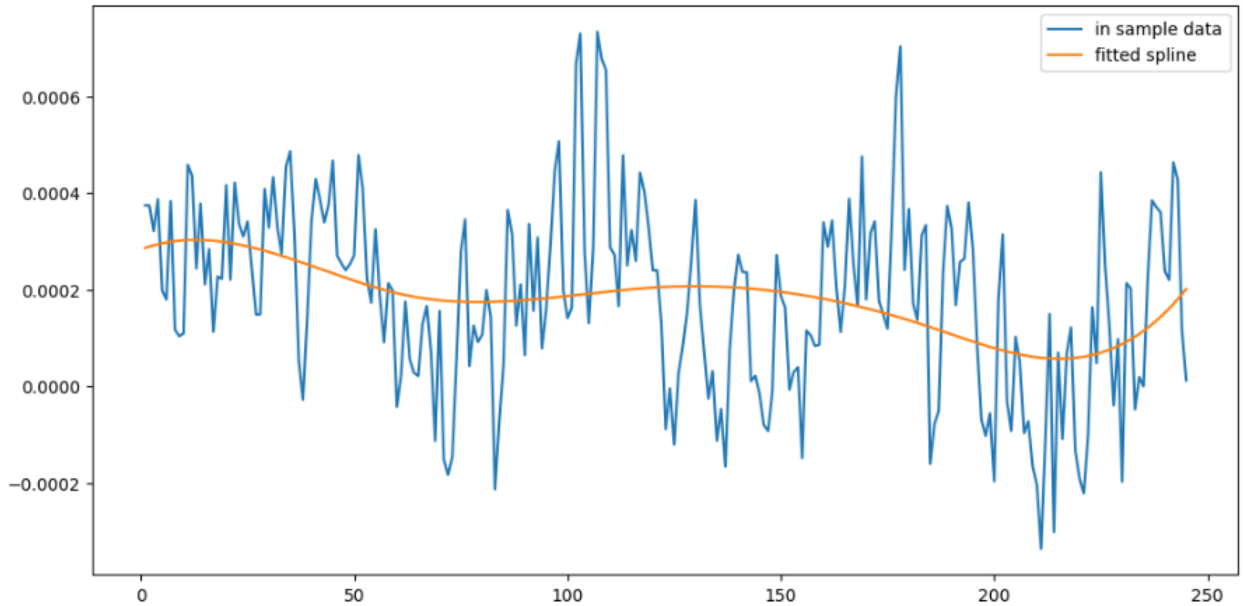


Figure 33: In Sample Spline for Alpha Weighting

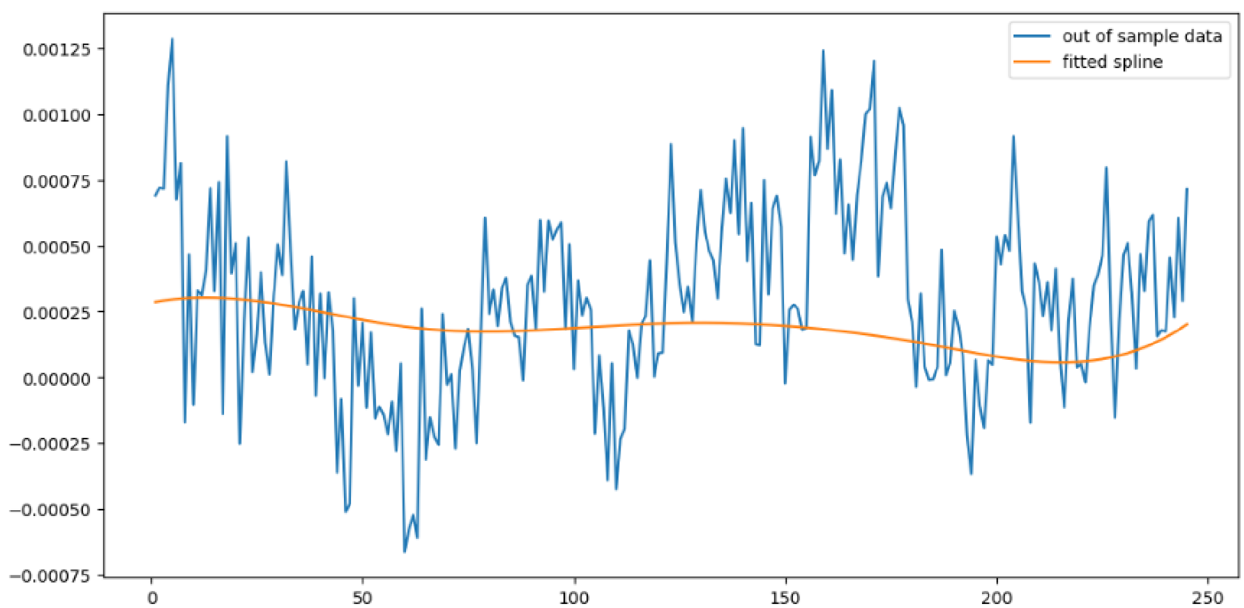


Figure 34: Fitted Spline Compared to Out of Sample Data

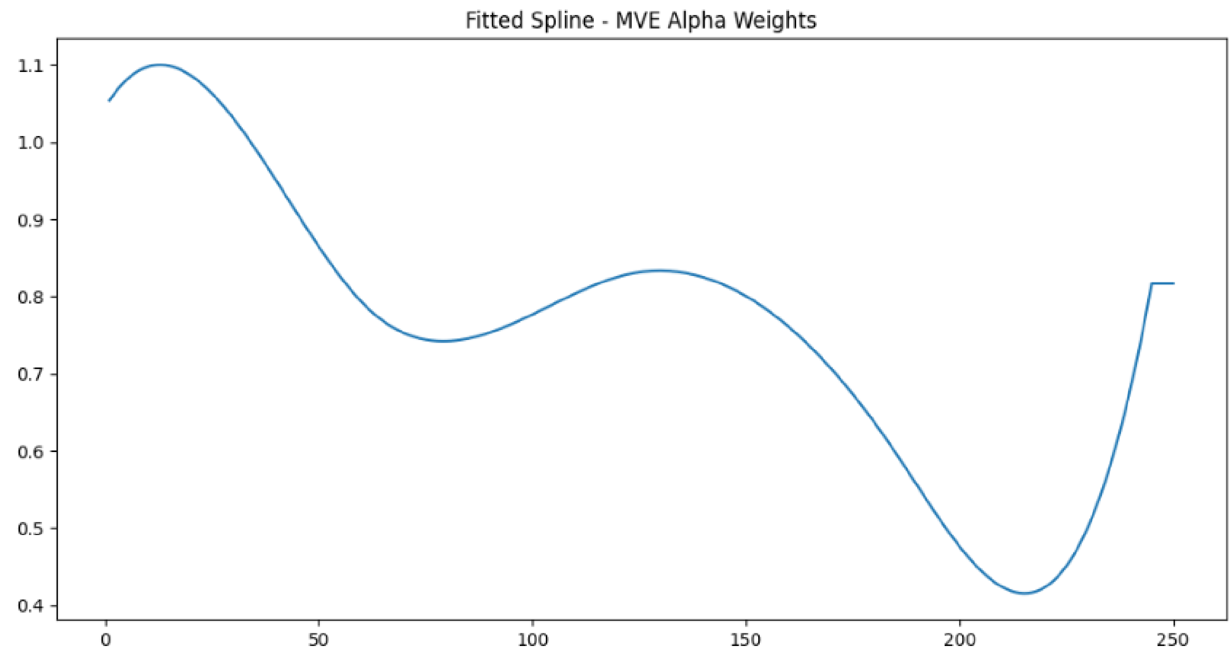


Figure 35: Shape of the Weight Multiplier by Relative Day